

Distributed Systems

Fault



CUHK

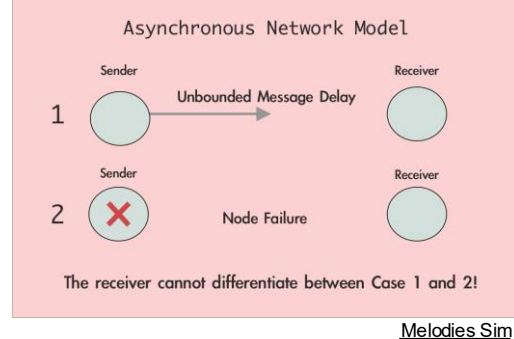
Assumptions so far

- E.g., assume no message lost
- E.g., assume each peer is honest
- Those assumptions might not hold in practice
- Lead to the field of **dependable computing**

Fault Model

- Crash Failure

- Permanently Dead vs. Slow vs Dead-and-Recovering
 - Under asynchronous model
 - Can't differentiate between slow network and node failure
 - Under synchronous model
 - Can be detected using a fault-detection method (e.g., time-out)
- On failure, we assume it **fail-stop** like:
 - 1) the program crashes
 - i.e., it fails like segmentation fault
 - Rule out the case of 'exception caught' in, say, Java
 - 2) others can detect its failure //if others can't accurately detect, it is **crash-stop**
 - 3) the internal state is lost (state in RAM lost) → hence, even **it is recovered, it is back to the initial state**
 - i.e., if it was a leader, even after it is recovered, it is back to the default non-leader state
 - Most protocols only deal with **fail-stop**
- A protocol that can't tolerate fail-stop won't be able to tolerate **arbitrary** crash-failure



Melodies Sim

Fault Model

- Crash Failure
- Network Failure
 - E.g., Network partition
 - We won't consider network partitioning until we reach CAP
- Byzantine Failure
 - Embrace every possible failure
 - *Equivocation*: M might forward you: $x=6$; me: $x=5$; barbie: $x=7$; kitty: $x=8$
 - M might not forward x out....
 - Due to various reasons like software bugs and **malicious, etc.**
 - But we seldom use a BFT protocol just to cater for software bugs
 - Too much overhead
 - Hence, when using BFT protocol, mostly for untrusted setting

How to detect crash-fault?

- Due to FLP, in asynchronous system, fault detection can't be both complete and accurate
 - Complete: you don't miss any faulty process
 - Accurate: you won't get any false alarm (i.e., a process is healthy but you regard it as fail because it is simply too slow)
- To a distributed protocol, which one is less evil?
 - We prefer complete over accurate
 - Then how to detect it? Fault-detection “algorithm”
 - Ping-ack (source-driven; demand-driven)
 - Heart-beating (target-driven)